LINMA2171 (2026–2027) — Numerical Analysis: Approximation, Interpolation, Integration

Homework 1

Deadline: Tuesday 3 November 2026 10:30am

Introduction. Many physical systems produce signals that can be modeled as a sum of a few complex exponentials. In this homework, we study *Prony's method*, a method that recovers exactly the parameters of an M-component exponential signal using uniform samples.
Specifically, we consider a signal modeled as an exponential sum of order M,

$$h(x) := \sum_{j=1}^{M} c_j \exp(\omega_j x), \quad x \geq 0. \tag{1}$$

with distinct numbers $\omega_j \in [-\alpha, 0] + i[-\pi, \pi)$ with $\alpha > 0$, and non-zero complex coefficients $c_j \neq 0$ for $j = 1, \ldots, M$. We assume that we have access to $2M$ uniform discrete measurements

$$h_k := h(k) = \sum_{j=1}^{M} c_j \exp(\omega_j k), \quad k = 0, \ldots, 2M-1. \tag{2}$$

The goal of this homework is to develop a method to recover exactly both $\{\omega_j\}_{j=1}^{M}$ and $\{c_j\}_{j=1}^{M}$ from these measurements, analyze its sensitivity, and apply it to the recovery of a piecewise constant function.

Prony's method. To recover the parameters, we first map the exponents ω_j to nodes on the complex plane:

$$z_j := \exp(\omega_j), \quad j = 1, \ldots, M. \tag{3}$$

We then consider the polynomial whose roots are the z_j's,

$$p(z) = \prod_{j=1}^{M} (z - z_j) = \sum_{k=0}^{M-1} p_k z^k + z^M, \quad z \in \mathbb{C} \tag{4}$$

with coefficients $p_k \in \mathbb{C}$ for $k = 0, \ldots, M-1$ and $p_M = 1$.

(1.a) Show, using definitions (2) and (4), that

$$\sum_{k=0}^{M} p_k h_{k+m} = 0, \quad \forall m = 0, \ldots, M-1. \tag{5}$$

(1.b) Re-express (5) as linear system

$$\mathbf{H}\mathbf{p} = \mathbf{b} \tag{6}$$

where $\mathbf{p} = [p_0, \ldots, p_{M-1}]^T \in \mathbb{C}^M$ is the unknown vector. Provide explicit expressions for the entries of $\mathbf{H}$ and $\mathbf{b}$, and clearly state their dimensions.
Hint: *The right hand side is non-zero* $\mathbf{b} \neq \mathbf{0}$.

(1.c) Define the following matrix

$$\mathbf{V} = (z_j^{k-1})_{k=1,j=1}^{M,M} \in \mathbb{C}^{M \times M}, \tag{7}$$

show that the matrix $\mathbf{H}$ can be factorized as

$$\mathbf{H} = \mathbf{V}\,\mathrm{diag}(\mathbf{c})\mathbf{V}^T, \tag{8}$$

where $\mathbf{c} = [c_1, \ldots, c_M]^T \in \mathbb{C}^M$.

(1.d) Using (8), obtain a closed form expression for the determinant of $\mathbf{H}$. Your expression should depend *solely* on $\{c_i\}_{i=1}^M$ and $\{z_j\}_{j=1}^M$. Given the *a priori* knowledge that the $\{h_k\}_{k=0}^{2M-1}$ are generated exactly according to (2) with distinct ω_j and $c_j \neq 0$, conclude that $\mathbf{H}$ is invertible.
Hint: *The matrix* $\mathbf{V}$ *is a Vandermonde matrix which possesses a closed form expression for its determinant.*

(1.e) Given $\mathbf{p} = [p_0, \ldots, p_{M-1}]$ describe how we can reconstruct the z_j's, and subsequently recover $\{c_j\}_{j=1}^M$. Motivate your construction and show that it is unique.

(1.f) Implement the above method. Your implementation should take as input a vector of measurements $\{h_k\}_{k=0}^{2M-1}$, where $k = 0, \ldots, 2M-1$, and it should output the recovered vectors $\{\omega_j\}_{j=1}^M$ and $\{c_j\}_{j=1}^M$.

Non-Uniform Sampling. For this experiment, we ask you to fix $\alpha = 1$, $M = 5$, and sample

$$c_j \sim \mathcal{N}(0,1), \qquad \omega_j \sim \mathcal{U}(-\alpha, 0) + i\mathcal{U}(-\pi/2, \pi/2), \quad \forall j.$$

The goal is to study the robustness of the method under non-uniform sampling.

(2.a) Generate the following measurements,

$$h_k^\epsilon := h(k + \epsilon_k) = \sum_{j=1}^M c_j \exp\Big(\omega_j(k + \epsilon_k)\Big),$$

where $\epsilon = (\epsilon_0, \ldots, \epsilon_{2M-1}) \in \mathbb{R}^{2M}$ is a sampling error whose components are drawn independently as $\epsilon_k \sim \mathcal{U}(-\eta, \eta)$ for a given $\eta > 0$. Apply your algorithm directly to $\{h_k^\epsilon\}_{k=0}^{2M-1}$ to obtain $\{\hat{c}_j\}_{j=1}^M$ and $\{\hat{\omega}_j\}_{j=1}^M$, and study the evolution of the continuous reconstruction error

$$\mathcal{E} = \frac{1}{N+1} \sum_{n=0}^N \left| h(x_n) - \sum_{j=1}^M \hat{c}_j \exp\Big(\hat{\omega}_j x_n\Big) \right|^2,$$

evaluated on a fine uniform grid $\{x_n\}_{n=0}^N \subset [0, 2M-1]$ with $N = 500$, for different perturbations $\eta \in \{10^{-9}, 10^{-6}, 10^{-3}, 10^{-1}\}$. Provide a visualization. What do you observe? Explain in which step of the method this inaccuracy causes a problem.

(2.b) To mitigate the effect of irregular sampling, consider a preprocessing step:

(i) Interpolate the non-uniform measurements $\{(k + \epsilon_k, h_k^\epsilon)\}_{k=0}^{2M-1}$ to construct an interpolant $\hat{h}(x)$.

(ii) Evaluate it at integer grid points $k = 0, \ldots, 2M-1$ to produce uniform measurements $\{\hat{h}(k)\}_{k=0}^{2M-1}$.

(iii) Apply Prony's method to these resampled values.

Choose and justify an interpolation scheme, implement it, and plot the resulting reconstruction error alongside the result of the previous question for $\eta \in \{10^{-9}, 10^{-6}, 10^{-3}, 10^{-1}\}$. Discuss whether this preprocessing actually resolves the issue, and explain how the interpolation error affects the rest of the algorithm.

Recovery of piecewise constant function. Let us consider a piecewise constant function given as

$$f(x) = \sum_{j=1}^M b_j \mathbf{1}_{[T_j, T_{j+1})}(x), \tag{9}$$

where $b_j \in \mathbb{R}\ \forall j$, $0 \leq T_1 < T_2 < \ldots < T_{M+1}$, and $\mathbf{1}_{[a,b)}$ denotes the characteristic function of $[a, b)$. In order to connect this problem to ours, let us take the Fourier transform of f

$$\mathcal{F}\{f\}(\gamma) := \hat{f}(\gamma) = \sum_{j=1}^M \frac{b_j}{i\gamma}(e^{-i\gamma T_j} - e^{-i\gamma T_{j+1}}) \tag{10}$$

(3.a) Assume you have access to K Fourier measurements $\{\hat{f}(k\Delta_\gamma)\}_{k=0}^{K-1}$. Re-express (10) in the same format as (2). You should clearly indicate what the unknowns are in the same formalism as (2).
Hint: *Your final answer's left hand side should read* $\hat{h}(\gamma) := (i\gamma)\hat{f}(\gamma)$. *Hence, in the formalism of* (2), *the measurements are* $\hat{h}_k := \hat{h}(k\Delta_\gamma) = (ik\Delta_\gamma)\hat{f}(k\Delta_\gamma)$.

(3.b) Give the minimal number of measurements K, and a condition on the step size Δ_γ such that you can perfectly recover $\{T_j\}_{j=1}^{M+1}$ and $\{b_j\}_{j=1}^{M}$. Justify your answer.
Hint: *For* Δ_γ, *look at the hypothesis on* ω_j *in the problem statement.*

(3.c) Set $M = 4$, you are free to choose $\{b_j\}_{j=1}^{4}$ and $\{T_j\}_{j=1}^{5}$ as you want. Thereafter, using (10), generate $\{\hat{f}(k\Delta_\gamma)\}_{k=0}^{K-1}$ where K and Δ_γ are chosen according to your previous answer. Verify that you correctly recover $\{b_j\}_{j=1}^{4}$ and $\{T_j\}_{j=1}^{5}$ using your approach.

(3.d) Keep the same $\{T_j\}_{j=1}^{5}$ and $\{b_j\}_{j=1}^{4}$ as before, and generate Fourier measurements using (10) for $K = 10, 20, 50, 100$. Approximate the inverse Fourier transform for each K, and compare these reconstructions alongside the fixed Prony reconstruction obtained in the previous question. What do you observe? Why does Prony's method succeed here with so few measurements while numerical integration struggles?
Hint: *For a real signal, the inverse Fourier transform of* $\hat{f}$ *reads*

$$f(x) = \frac{1}{\pi}\int_0^\infty \Re\{\hat{f}(\gamma)e^{i\gamma x}\}d\gamma.$$

Given our measurements $\{\hat{f}(k\Delta_\gamma)\}_{k=0}^{K-1}$, *we will approximate it by first truncating the integral up to* $(K-1)\Delta_\gamma$

$$f(x) \approx \frac{1}{\pi}\int_0^{(K-1)\Delta_\gamma} \Re\{\hat{f}(\gamma)e^{i\gamma x}\}d\gamma,$$

and subsequently performing a trapezoidal approximation *of this integral, that is,*

$$f(x) \approx \frac{\Re\{\hat{f}(0)\}\Delta_\gamma}{2\pi} + \frac{\Delta_\gamma}{\pi}\sum_{k=1}^{K-2} \Re\{\hat{f}(k\Delta_\gamma)e^{i\Delta_\gamma kx}\} + \frac{\Delta_\gamma}{2\pi}\Re\{\hat{f}((K-1)\Delta_\gamma)e^{i\Delta_\gamma(K-1)x}\}.$$

You can use this formula to compare the reconstructions.

Practical information.

Meetings with the TA: By appointment.
Writing : You must do all the writing (report and code) ***individually***. Never share your production, but you are allowed, and even encouraged, to exchange ideas on how to address the homework. If you use an LLM, mention it in the report and explain for which purpose it was used. However, we believe that this assignment will be more useful to you—and the report more pleasant to read for us—if you use LLMs sparingly, if at all.
Questions: Feel free to contact me at theo.hanon@uclouvain.be to ask questions or to set up a meeting.
Submission: Using Moodle. Keep in mind that the deadline is automatically enforced, using the clock of Moodle!
Python/Octave/Matlab/Julia codes: Please don't copy your code to your report and add the `.py/.ipynb/.m/.jl` files as attachments. Add a script that reproduces all your results.
Language: Reports in French are accepted without penalty. However, English is strongly encouraged. The quality of the English will not impact the grade, provided that the text is intelligible.

Deadline: 3 November, 2026 at 10:30am

References

[1] Gerlind Plonka and Manfred Tasche. “Prony methods for recovery of structured functions”. en. In: *GAMM-Mitteilungen* 37.2 (2014), pp. 239–258. ISSN: 0936-7195, 1522-2608. DOI: 10.1002/gamm.201410011. URL: https://onlinelibrary.wiley.com/doi/10.1002/gamm.201410011.